

Package #166

Integrate a passive double-walled sealed sheath around the tunneled hemodialysis catheter to create a permanently closed conduit, eliminating external fluid ingress without active valves.

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'patency variability' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters show conflicting patency outcomes across studies?". [fda_maude] 100 records retrieved live (1992-2070). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 1000009: Injury [SLIP-CATH BEACON TIP CATHETER]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'Comprehensive management of hemodialysis catheter-related bloodstream infections: a narrative review.'; 'Central line-associated thrombosis and infections in patients with acute leukemias: The chicken or the egg dilemma.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Closed-system technology that physically isolates the catheter lumen from external contaminants, thereby reducing contamination risk.

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.